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Research Interests

Wireless networks, Internet of Things, security and privacy, machine learning, edge computing.

Education

University of Florida Ph.D. in Electrical and Computer Engineering <i>Advisor: Dr. Yuguang “Michael” Fang</i>	Gainesville, FL August 2016
Tsinghua University M.S. in Electronic Engineering	Beijing, China July 2012
Huazhong University of Science and Technology B.E. in Electronics and Information Engineering <i>Special Class for the Gifted Young</i>	Wuhan, China July 2009

Employment

Assistant Professor Department of Electrical and Computer Engineering The University of Texas at San Antonio, San Antonio, TX	08/2019 – present
Assistant Professor School of Electrical and Computer Engineering Oklahoma State University, Stillwater, OK	08/2016 – 07/2019

Honors and Awards

NSF CAREER Award	2021
Rising Star in Networking and Communications, N2Women	2021
IEEE INFOCOM Distinguished TPC Member, IEEE Communications Society	2020
NSF CRII Award	2019
UT Rising STARS Award	2019
President’s Cup Award, Oklahoma State University	2018
IEEE GLOBECOM Best Paper Award	2017
Outstanding Achievement Award, University of Florida	2012, 2013
Guanghua Scholarship for Excellent Students, Tsinghua University	2011
Outstanding Graduate Award, Huazhong University of Science and Technology	2009

Travel Grants

Travel Grant for NSF CSR PI Meeting, Seattle, WA	2018
Travel Grant for Hands-on Labs for Security Education Workshop, Syracuse, NY	2018
Travel Grant for IEEE International Conference on Communications, Kansas City, MO	2018
Travel Grant for NSF CSR Aspiring PIs Workshop, Alexandria, VA	2018
Travel Grant for CyberW Workshop and ACM CCS, CRA-W	2017
USENIX Security’17 Travel Grant for Women, Google	2017
Travel Grant for the CRA-W Early Career Mentoring Workshop, NSF	2016
Travel Grant from the South Big Data Hub, NSF	2016
IEEE CNS Travel Grant for Junior Faculty, Army Research Office	2016, 2017
ACM MobiHoc Student Travel Grant, NSF	2015
IEEE GLOBECOM Student Travel Grant, IEEE Communication Society	2014, 2015

Publications

Peer-Reviewed Journal Articles

1. X. Dong, Y. Zhang, Y. Guo, Y. Gong, Y. Shen, and J. Ma, “PRAM: a Practical Sybil-Proof Auction Mechanism for Dynamic Spectrum Access with Untruthful Attackers,” accepted to appear in *IEEE Transactions on Mobile Computing*, DOI:10.1109/TMC.2021.3090103.
2. R. Hu, Y. Guo, and Y. Gong, “Concentrated Differentially Private Federated Learning with Performance Analysis,” *IEEE Open Journal of the Computer Society*, pp. 276-289, 2021, doi: 10.1109/OJCS.2021.3099108.
3. R. Hu, Y. Guo, H. Li, Q. Pei, and Y. Gong, “Personalized Federated Learning with Differential Privacy,” *IEEE Internet of Things Journal*, vol. 7, no. 10, 2020.
4. G. Zhang, H. Qin, Y. Ke, J. Chen, and Y. Gong, “Phased Groupwise Face Alignment”, *IEEE Access*, vol. 8, pp. 62415-62422, 2020, doi: 10.1109/ACCESS.2020.2983722.
5. G. Zhang, L. Pan, J. Chen, Y. Gong, and F. Liu, “Joint Alignment of Image Faces”, *IEEE Access*, vol. 8, pp. 114884-114891, 2020, doi:10.1109/ACCESS.2020.3003332.
6. Z. Yu, Y. Gong, S. Gong, and Y. Guo, “Joint Task Offloading and Resource Allocation in UAV-Enabled Mobile Edge Computing”, *IEEE Internet of Things Journal*, vol. 7, no. 4, pp. 3147-3159, 2020.
7. Z. Huang, R. Hu, Y. Guo, E. Chan-Tin, and Y. Gong, “DP-ADMM: ADMM-based Distributed Learning with Differential Privacy”, *IEEE Transactions on Information Forensics & Security*, July 2019, doi: 10.1109/TIFS.2019.2931068.
8. W. Cui, J. Yu, Y. Gong, and E. Chan-Tin, “Efficient, Effective, and Realistic Website Fingerprinting Mitigation”, *EAI Transactions on Security and Safety*, vol. 6, no. 20, April, 2019, doi: 10.4108/eai.29-1-2019.161977.
9. S. Errapotu, J. Wang, Y. Gong, J. Cho, M. Pan, and Z. Han, “SAFE: Secure Appliance Scheduling for Flexible and Efficient Energy Consumption for Smart Home IoT”, *IEEE Internet of Things Journal*, vol. 5, no. 6, pp. 4380 - 4391, Aug. 2018, doi: 10.1109/JIOT.2018.2866998.
10. J. Wang, S. Errapotu, Y. Gong, L. Qian, R. Jäntti, M. Pan, and Z. Han, “Data-Driven Optimization Based Primary Users’ Operational Privacy Preservation”, *IEEE Transactions on Cognitive Communications and Networking*, vol. 4, no. 2, pp. 357-367, May 2018.
11. Y. Guo, M. Pan, and Y. Gong, “Aggregation-Based Colocation Datacenter Energy Management in Wholesale Markets”, *IEEE Transactions on Cloud Computing*, May 2018, doi: 10.1109/TCC.2018.2836424.

12. X. Dong, Y. Gong, J. Ma, and Y. Guo, "Protecting Operation-Time Privacy of Primary Users in Downlink Cognitive Two-Tier Networks", *IEEE Transactions on Vehicular Technology*, vol. 67, no. 7, pp. 6561-6572, July 2018.
13. Y. Gong, C. Zhang, Y. Fang, and J. Sun, "Protecting Location Privacy for Task Allocation in Ad-Hoc Mobile Cloud Computing", *IEEE Transactions on Emerging Topics in Computing*, vol. 6, no. 1, pp. 110-121, Jan.-March 2018.
14. Y. Guo, M. Pan, Y. Gong, and Y. Fang, "Dynamic Multi-Tenant Coordination for Sustainable Colocation Data Centers", *IEEE Transactions on Cloud Computing*, vol. 7, no. 3, July-Sept. 2019.
15. Y. Gong, L. Wei, Y. Guo, C. Zhang, and Y. Fang, "Optimal Task Recommendation for Mobile Crowdsourcing with Privacy Control", *IEEE Internet of Things Journal*, vol. 3, no. 5, pp. 745-756, Oct. 2016.
16. Y. Gong, Y. Fang, and Y. Guo, "Private Data Analytics on Biomedical Sensing Data Via Distributed Computation", *IEEE/ACM Transactions on Computational Biology and Bioinformatics*, vol. 13, no. 3, pp. 431-444, May/June 2016.
17. Y. Gong, Y. Cai, Y. Guo, and Y. Fang, "A Privacy-Preserving Scheme for Incentive-Based Demand Response in the Smart Grid", *IEEE Transactions on Smart Grid*, vol. 7, no. 3, pp. 1304-1313, May 2016.
18. Y. Guo, Y. Gong, Y. Fang, P. Khargonekar, and X. Geng, "Energy and Network Aware Workload Management for Sustainable Data Centers with Thermal Storage", *IEEE Transactions on Parallel and Distributed Systems*, Vol. 25, No. 8, pp. 2030-2042, August 2014.

Conference Proceedings

1. R. Hu, Y. Gong, and Y. Guo, "Federated Learning with Sparsification-Amplified Privacy and Adaptive Optimization", *IJCAI 2021*, Accepted to Appear. (**Acceptance Ratio: 13.9%**)
2. S. Amin, L. Li, Y. Guo, M. Pan, and Y. Gong, "Geo-Indistinguishability for Crowdsourced-Based Radio Environment Map Construction", *IEEE GLOBECOM*, Taipei, Taiwan, Dec. 7-11, 2020.
3. R. Hu and Y. Gong, "Trading Data For Learning: Incentive Mechanism For On-Device Federated Learning", *IEEE GLOBECOM*, Taipei, Taiwan, Dec. 7-11, 2020.
4. Z. Gao, R. Hu, and Y. Gong, "Certified Robustness of Graph Classification against Topology Attack with Randomized Smoothing", *IEEE GLOBECOM*, Taipei, Taiwan, Dec. 7-11, 2020.
5. P. Bhatt, Y. Guo, and Y. Gong, "Discounting Lives: Analyzing the causes of Hispanic Adversity raging in the Healthcare sector during Covid-19 using Cumulative Inequality Theory", *Hawaii International Conference on System Sciences*, Virtual Venue, Jan. 5-8, 2021.
6. R. Chen, L. Li, J. Chen, R. Hou, Y. Gong, Y. Guo, and M. Pan, "COVID-19 Vulnerability Map Construction via Location Privacy Preserving Mobile Crowdsourcing", *IEEE GLOBECOM*, Taipei, Taiwan, Dec. 7-11, 2020.
7. R. Hu, Y. Guo, H. Li, Q. Pei, and Y. Gong, "Privacy-Preserving Personalized Federated Learning", *IEEE ICC*, Dublin, Ireland, June 7-11, 2020.
8. R. Hu, Y. Guo, M. Pan, and Y. Gong, "Targeted Poisoning Attacks on Social Recommender Systems", *IEEE GLOBECOM*, Hawaii, Dec. 9-13, 2019.
9. Z. Huang, M. Pan, and Y. Gong, "Robust Truth Discovery Against Data Poisoning in Mobile Crowdsensing", *IEEE GLOBECOM*, Hawaii, Dec. 9-13, 2019.
10. R. Li, L. Wang, Y. Gong, M. Song, M. Pan, and Z. Han, "Dynamic Cache Placement, Node Association, and Power Allocation in Fog Aided Networks", *IEEE GLOBECOM*, Hawaii, Dec. 9-13, 2019.

11. J. Ding, S. Errapotu, H. Zhang, Y. Gong, M. Pan and Z. Han, "Stochastic ADMM Based Distributed Machine Learning with Differential Privacy", *EAI SecureComm*, Orlando, Oct. 23-25, 2019. (Acceptance Ratio: 25.5%)
12. Y. Guo and Y. Gong, "Practical Collaborative Learning for Crowdsensing in the Internet of Things with Differential Privacy", *IEEE CNS*, Beijing, China, May 30 - June 1, 2018. (Acceptance Ratio: 27.8%)
13. Y. Hu, X. Li, J. Liu, H. Ding, Y. Gong, and Y. Fang, "Mitigating Traffic Analysis Attack in Smartphones with Edge Network Assistance", *IEEE ICC*, Kansas City, MO, May 20-24, 2018.
14. W. Cui, J. Yu, Y. Gong, and E. Chan-Tin, "Realistic Cover Traffic to Mitigate Website Fingerprinting Attacks", *IEEE ICDCS workshop*, Vienna, Austria, July 2-5, 2018.
15. J. Liu, Y. Hu, H. Yue, Y. Gong, Y. Fang, "A Cloud-Based Secure and Privacy-Preserving Clustering Analysis of Infectious Disease", *IEEE Symposium on Privacy-Aware Computing (PAC)*, Washington, DC, 2018.
16. Y. Guo, Y. Gong, L. Njilla, and C. Kamhoua, "A Stochastic Game Approach to Cyber-Physical Security with Applications to Smart Grid", *IEEE INFOCOM workshop*, Honolulu, HI, April 15-19, 2018.
17. R. Hu and Y. Gong, "Secret Dispersion: Secure Data Delivery in Cyber Physical System", *IEEE CNS poster*, Beijing, China, May 30 - June 1, 2018.
18. J. Wang, Y. Gong, L. Qian, R. Jäntti, M. Pan, and Z. Han, "Primary Users' Operational Privacy Preservation via Data-Driven Optimization", *IEEE GLOBECOM*, Singapore, December 4-8, 2017. **(Best Paper Award)**
19. Z. Huang and Y. Gong, "Differential Location Privacy for Crowdsourced Spectrum Sensing", *IEEE CNS*, Las Vegas, NV, October 9-11, 2017. (Acceptance Ratio: 29.9%)
20. Y. Gong, C. Zhang, Y. Hu, and Y. Fang, "Privacy-Preserving Genome-Aware Remote Health Monitoring", *IEEE GLOBECOM*, Washington, D.C., December 4-8, 2016.
21. L. Wei, C. Zhang, Y. Gong, Y. Fang, and K. Chen, "A Firewall of Two Clouds: Preserving Outsourced Firewall Policy Confidentiality with Heterogeneity", *IEEE GLOBECOM*, Washington, D.C., December 4-8, 2016.
22. Y. Gong, Y. Fang, and Y. Guo, "Privacy-Preserving Collaborative Learning for Mobile Health Monitoring", *IEEE GLOBECOM*, San Diego, CA, December 6-10, 2015.
23. Y. Gong, Y. Guo, and Y. Fang, "A Privacy-Preserving Task Recommendation Framework for Mobile Crowdsourcing", *IEEE GLOBECOM*, Austin, TX, December 8-12, 2014. (selected as **"Best of IEEE GLOBECOM"**, 50 among 1000 accepted papers)
24. Y. Guo, Y. Gong, Y. Fang, and P. Khargonekar, "Stochastic Minimization of Imbalance Cost for a Virtual Power Plant in Electricity Markets", *IEEE ISGT*, Washington, DC, February 19-22, 2014.
25. Y. Guo, Y. Gong, Y. Fang, P. Khargonekar, and X. Geng, "Optimal Power and Workload Management for Green Data Centers with Thermal Storage", *IEEE GLOBECOM*, Atlanta, GA, December 9-13, 2013.
26. Y. Gong, J. Chen and G. Su, "Quasi-Convex Optimization of Metrics in Biometric Score Fusion", *ICIG*, Hefei, China, August 12-15, 2011.

Research Funding

"CAREER: Ubiquitous and Time-Critical Federated Learning with Cooperative Mobile Edge Networking", National Science Foundation, \$509,014, 10/01/2021 - 09/30/2026, Sole PI.

“Collaborative Research: CNS Core: Medium: Towards Federated Learning over 5G Mobile Devices: High Efficiency, Low Latency, and Good Privacy”, National Science Foundation, \$250,000, 10/01/2021 - 09/30/2025, Co-PI.

“GREAT: Regional Natural Hazard Risk Assessment Methodology toward Resilient 5G Telecommunication Infrastructure Deployment”, the Office of the Vice President for Research, Economic Development, and Knowledge Enterprise (VPREDKE) at UTSA, \$20,000, 10/01/2021 - 07/31/2022, Co-PI.

“A Flexible Testbed for Cyber Deception, Cyber Hardening, and Disinformation Research”, Department of Defense, \$472,921, 07/01/2021 - 06/30/2022, Co-PI.

“RAPID: Collaborative: Location Privacy Preserving COVID-19 Symptom Map Construction via Mobile Crowdsourcing for Proactive Constrained Resource Allocation”, National Science Foundation, \$100,000, 05/04/2020 - 05/03/2022, Co-PI.

“REU: CRII: NeTS: Embracing Dynamic Spectrum Sharing without Privacy Concerns”, National Science Foundation, \$16,000, 10/01/2020 - 09/30/2022, Sole PI.

“CRII: NeTS: Embracing Dynamic Spectrum Sharing without Privacy Concerns”, National Science Foundation, \$175,000, 10/01/2019 - 09/30/2022, Sole PI.

“Association of Cyber and Visual IDs of Drones with Backscatter Tags”, Air Force Research Laboratory, \$10,000, 10/01/2019 - 01/01/2020, Sole PI.

“Secure Wireless Communication against Spoofing Attacks in UAV Systems”, Air Force Research Laboratory, \$17,040, 05/15/2019 - 07/15/2019, Sole PI.

Teaching

Instructor

<i>Data Science and Cybersecurity</i> (EE 5263) The University of Texas at San Antonio	Spring 2021
<i>Introduction to Computer Networks</i> (EE 4673/EE 5283) The University of Texas at San Antonio	Fall 2019
<i>Computer Networks</i> (ECEN/CS 4283) Oklahoma State University	Spring 2017, Spring 2018
<i>Computer Based Systems</i> (ENSC 3213) Oklahoma State University	Fall 2018
<i>Security and Privacy in Big Data</i> (ECEN 5060) Oklahoma State University	Fall 2016, Fall 2017

Advising

Ph.D. Students, Primary Advisor

Ms. Rui Hu (Fall 2017 - present)
Mr. Zhidong Gao (Fall 2019 - present)
Mr. Yu Zhang (Fall 2019 - present)
Mr. Zhenxiao Zhang (Fall 2020 - present)

M.S. Student

Mr. John Lopez (Spring 2020 - present)

Mr. Domnic Eaton (Fall 2017 - Fall 2018)

Mr. Zonghao Huang (Spring 2017 - Spring 2019)

Graduate Advisory Committee

Adnan Menon, MSEE (OSU), Advisor: Dr. Subhash Kak

Rakshitha Ravula, MSEE (OSU), Advisor: Dr. Subhash Kak

Susruth Tatavarthy, MSEE (OSU), Advisor: Dr. James Stine

Prasanna Pale, MSCS (OSU), Advisor: Dr. Eric Chan-Tin

Nikhil Jain, MSCS (OSU), Advisor: Dr. Eric Chan-Tin

Mohammad Hossein Khaleghian (OSU), Advisor: Dr. Yongwei Shan

Md. Farhan Haque, PhD (UTSA), Advisor: Dr. Ram Krishnan

Yuvaraj Munlan, PhD (UTSA), Advisor: Dr. Miltos Alamaniotis

Rahul Mundlamuri, PhD (UTSA) Advisor: Dr. David Akopian

Devasena Inupakutika, PhD (UTSA) Advisor: Dr. David Akopian

Randy Klepetko, PhD (UTSA), Advisor: Dr. Ram Krishnan

Undergraduate Students

Mr. Talha Khan

Mr. Erick Galicia

Mr. Andy Hernandez

Mr. Mario Alday

Ms. Kisa Fors-Francis

Professional Services

Journal Editorship

Editor, IEEE Wireless Communications, 2019 – present

Conference Organizing

2019: Organizing Committee, UTSA AI Summit

2019: Publication Chair, 16th EAI International Conference on Mobile and Ubiquitous Systems: Computing, Networking and Services (MobiQuitous)

2017: Publicity Chair, IEEE International Conference on Mobile Ad hoc and Sensor Systems (MASS)

Panel Reviewers

2017 – 2019, 2021: NSF Panelist

Conference Technical Program Committee Members

2017 – 2021: IEEE International Conference on Computer Communications (INFOCOM)

2017, 2018, 2021: IEEE Conference on Communications and Network Security (CNS)

2020: International Joint Conference on Artificial Intelligence (IJCAI)
2018: IEEE International Conference on Communications (ICC)
2017: IEEE/CIC International Conference on Communications in China (ICCC)
2017: The 7th International Workshop on Internet of Things: Privacy, Security and Trust (IoTPST)
2017: International Conference on Industrial, Engineering & Other Applications of Applied Intelligent Systems (IEA/AIE)
2015: IEEE International Conference on Computer and Information Technology (CIT)

Journal Reviewers

IEEE Transactions on Neural Networks and Learning Systems
IEEE Internet of Things Journal
IEEE Transactions on Parallel and Distributed Systems
IEEE Transactions on Services Computing
IEEE Transactions on Information Forensics & Security
IEEE Transactions on Mobile Computing
ACM Computing Surveys
Transportation Research Part C: Emerging Technologies
IEEE Transactions on Vehicular Technology
IEEE Transactions on Emerging Topics in Computing
IEEE Wireless Communications
IEEE Network
IEEE Transactions on Smart Grid
IEEE Journal on Selected Areas in Communications

Professional Memberships

Senior Member, Institute of Electrical and Electronics Engineers (IEEE)
Member, IEEE Communication Society (IEEE ComSoc)
Member, Networking Networking Women (N² Women)

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